

ENTITY ANALYSIS

Chapter -2

2.Introduction:

The **AJIO E-Commerce Database Management System** is designed to efficiently manage the large amount of data involved in an online shopping platform. It stores and organizes information related to customers, products, categories, orders, payments, deliveries, and sellers in a structured database. A Database Management System (DBMS) enables fast data retrieval, secure storage, accurate transaction processing, and efficient inventory management. It also minimizes data redundancy, maintains data integrity, and supports multiple users accessing the system simultaneously. By using entities, attributes, primary keys, foreign keys, and relationships, the database ensures smooth communication between different components of the e-commerce platform. This well-structured database improves operational efficiency, enhances the customer shopping experience, and supports reliable decision-making for the business.

2.1 ENTITY ATTRIBUTES

Entity	Description
Customer	Stores customer details and account information.
Product	Stores product information available for sale.
Category	Stores product category details.
Order	Stores customer order information.
Order Item	Stores products included in each order.
Payment	Stores payment transaction details.
Delivery	Stores shipment and delivery information.
Seller	Stores seller and supplier details.
Review	Stores customer product reviews and ratings.
Wishlist	Stores products saved by customers for future purchase.

2.2 CUSTOMER TABLE

Attribute	Description
Customer_ID	Unique customer ID
First_Name	Customer's first name
Last_Name	Customer's last name
Email	Customer email address
Phone_Number	Customer contact number
Password	Customer account password
Gender	Customer gender

Date_of_Birth	Customer date of birth
Address	Customer delivery address
City	Customer city
State	Customer state
Pincode	Customer postal code
Registration_Date	Date of account creation

2.3 PRODUCT TABLE

Attribute	Description
Product_ID	Unique product ID
Product_Name	Product name
Brand	Product brand
Category_ID	Product category ID
Description	Product description
Price	Product price
Discount	Product discount
Stock_Quantity	Available stock
Size	Product size
Color	Product color
Image_URL	Product image
Rating	Customer rating

2.4 Category table :

Attribute	Description
Category_ID	Unique category ID
Category_Name	Category name
Category_Description	Category description

2.5 Order table :

Attribute	Description
Order_ID	Unique order ID
Customer_ID	Customer ID
Order_Date	Date of order
Total_Amount	Total order amount
Order_Status	Current order status
Shipping_Address	Delivery address

Delivery_Date	Expected delivery date
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2.6 Order item table :

Attribute	Description
Order_Detail_ID	Unique order item ID
Order_ID	Order ID
Product_ID	Product ID
Quantity	Quantity ordered
Unit_Price	Price per product
Total_Price	Total price

2.7 Payment table :

Attribute	Description
Payment_ID	Unique payment ID
Order_ID	Order ID
Payment_Method	Payment method
Payment_Date	Date of payment
Payment_Status	Payment status
Amount	Amount paid
Transaction_ID	Payment transaction ID

2.8 Delivery table :

Attribute	Description
Delivery_ID	Unique delivery ID
Order_ID	Order ID
Delivery_Partner	Delivery service provider
Tracking_Number	Shipment tracking number
Delivery_Status	Delivery status
Expected_Delivery_Date	Expected delivery date
Actual_Delivery_Date	Actual delivery date

2.9 Seller table :

Attribute	Description
Seller_ID	Unique seller ID
Seller_Name	Seller name

Email	Seller email
Phone_Number	Seller contact number
Address	Seller address
GST_Number	Seller GST number

2.10 Review table :

Attribute	Description
Review_ID	Unique review ID
Customer_ID	Customer ID
Product_ID	Product ID
Rating	Product rating
Review_Text	Customer review
Review_Date	Date of review

2.11 Wishlist table

Attribute	Description
Wishlist_ID	Unique wishlist ID
Customer_ID	Customer ID
Product_ID	Product ID
Date_Added	Date product added to wishlist

2.12 Relationships Required for the AJIO E-Commerce Database Management System

Relationship	Type	Description
Customer → Order	One-to-Many (1:M)	One customer can place multiple orders, but each order belongs to one customer.
Customer → Cart	One-to-One (1:1)	Each customer has one shopping cart.
Customer → Wishlist	One-to-One (1:1)	Each customer has one wishlist to save products.
Customer → Review	One-to-Many (1:M)	One customer can write multiple product reviews.
Category → Product	One-to-Many (1:M)	One category contains multiple products, while each product belongs to one category.
Seller → Product	One-to-Many (1:M)	One seller can sell many products.
Order → Order Item	One-to-Many (1:M)	One order contains one or more order items.
Product → Order Item	One-to-Many (1:M)	One product can appear in many order items.

Cart → Product	Many-to-Many (M:N)	A cart can contain many products, and a product can be added to many carts.
Wishlist → Product	Many-to-Many (M:N)	A wishlist can contain multiple products, and a product can appear in multiple wishlists.
Order → Payment	One-to-One (1:1)	Each order has one payment record.
Order → Delivery	One-to-One (1:1)	Each order has one delivery record.
Product → Review	One-to-Many (1:M)	One product can receive reviews from many customers.

2.13 Conclusion

The **AJIO E-Commerce Database Management System** provides a structured and efficient way to manage customer, product, order, payment, delivery, and seller information. By identifying the required entities, attributes, primary keys, and relationships, the database ensures accurate data storage, reduces redundancy, and maintains data integrity. The system supports secure transactions, efficient inventory management, and fast retrieval of information, enabling smooth online shopping operations. Overall, the database design improves system performance, enhances the customer experience, and provides a reliable foundation for the successful operation of an e-commerce platform.